Chemical Biology

ELIXIR CZECH REPUBLIC

Chemical entities of biological interest – such as small molecule metabolites – are essential building blocks of biological systems with crucial roles in human health and disease. Structural classification of these chemical entities is the first step to understanding these roles, but the number and complexity of known and possible chemical structures make any manual classification effort unfeasible. The chemoinformatics characterization of small molecules enhances their utilization in other molecular biology fields.
To effectively utilize large databases of chemical entities, users need well-designed interfaces that guarantee the interoperability of services and seamless processing of chemical data. This is especially important for analyses that concern, e.g., drug design and ligand docking, which should employ robust, reproducible and well-benchmarked screening methods.

Key Domains

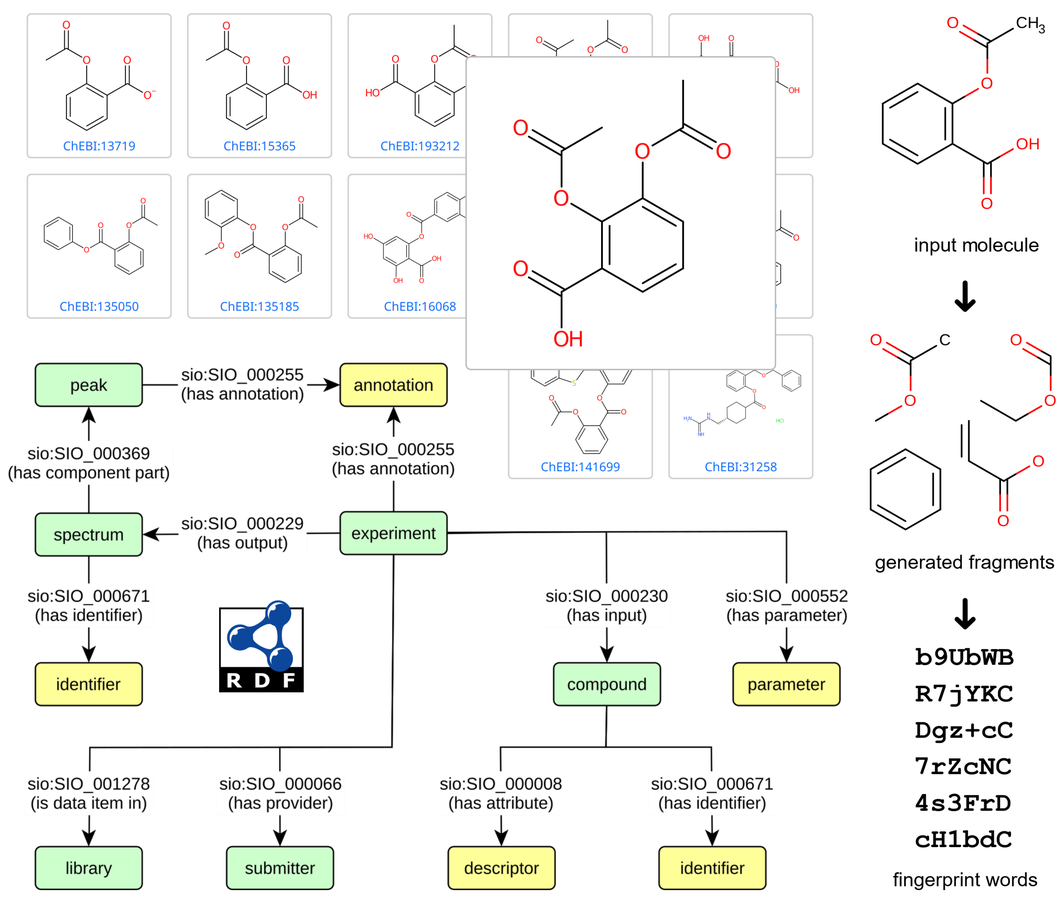
- development of interoperable services at a global scale
- development of tools for classification and characterisation of chemical entities

Strategic Goals

- integration of chemical datasets into one service
- interoperability with other ELIXIR Core Data Resources through the SPARQL protocol
- seamless extraction of useful information for computational methods at the interface of proteomics and cheminformatics
- application of new results to workflows for automated drug design utilizing chemical information

Semantic Web for FAIR Data

- design and implementation of life-science ontologies
- development of an in-house extended SPARQL engine
- integration of compound search services into SPARQL
- support for federated queries across multiple resources



Database Services & Tools

idsm.elixir-czech.cz
integrated database of small molecules

molmedb.upol.cz
membrane interactions of small molecules

channelsdb2.biodata.ceitec.cz
biomacromolecule channels, pores & tunnels

molmedb.upol.cz/lab
permeation calculations with COSMOperm

acc2.ncbr.muni.cz
fast calculation of partial atomic charges

spci.imtm.cz
interpretation of QSAR models

alphacharges.ncbr.muni.cz
pdbcharges.biodata.ceitec.cz
web servers for charges of proteins

ELIXIR Platforms & Communities

Data Platform
aims to identify key data resources across Europe and support the linkages between data and literature

Interoperability Platform
develops and encourages the adoption of standards to describe life science data

Metabolomics Community
is the study of small molecule metabolites in the metabolome

Collaboration

- **design of MolMeDB RDF model** (IOCB & UPOL)
- **charge calcs and drug design workflow** (MU, UPOL & SIB)
- **design of protein thermostability** (UWB & IOCB)
- **enhancing federated queries** (IOCB, SIB & UGent)

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